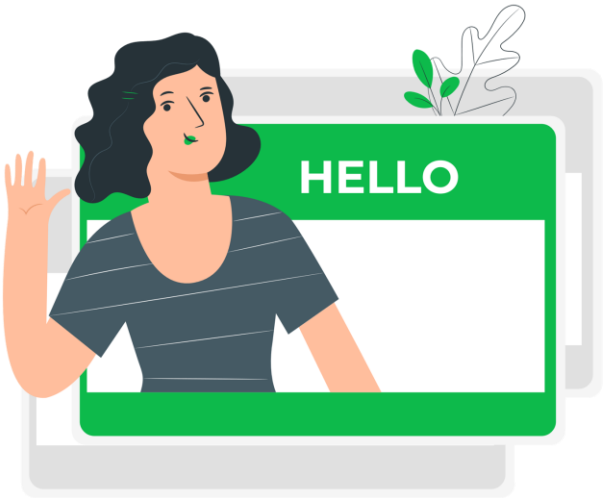
Data Structures in Python

GUVI | HCL

Skill Up. Level Up

Agenda

1. Introduction to Programming Basics
2. Software Installation & Environment Setup
3. Git & GitHub Basics
4. Creating and Running Python Programs
5. Python Data Types
6. Operators in Python
7. Conditional Statements
8. Scenario-Based Problems



Unit Overview

This unit is designed to build a **strong foundation in programming using Python**, while simultaneously introducing learners to **professional development practices using Git and GitHub**. By the end of the session, learners will not only understand basic programming concepts but also know how to manage and version their code like industry professionals.

What is Programming?

- Programming is the process of writing instructions for a computer
- Instructions are written using programming languages

Example

- ATM withdrawal process
- Online order placement

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What is Python?

- High-level, interpreted language
- Easy syntax, beginner-friendly
- Used in:
  - Data Science
  - Web Development
  - Automation
  - AI & ML

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Program vs Algorithm

Algorithm

- Step-by-step solution to a problem

Program

- Algorithm written using a programming language

Example Algorithm

1. Read two numbers
2. Add them
3. Display result

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IDE Setup (VS Code / PyCharm)

VS Code:

- Install Python extension
- Select interpreter

PyCharm:

- Create project
- Choose virtual environment

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Installing Python

- Download Python from official site
- Select "Add Python to PATH"
- Verify installation

Command

```
python --version
```


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Why Git is Important?

- Tracks code history
- Enables rollback
- Team collaboration

Example:

Undo accidental deletion

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Git vs GitHub

Feature	Git	GitHub
Type	Version Control Tool	Web-based Hosting Platform
Location	Local system (installed)	Cloud-based (hosted online)
Purpose	Track and manage code changes	Store, share, and collaborate on code
Access	Command line / GUI tools	Web interface + Git integration
Internet	Not required (works offline)	Required for remote access
Ownership	Developed by Linus Torvalds	Owned by Microsoft

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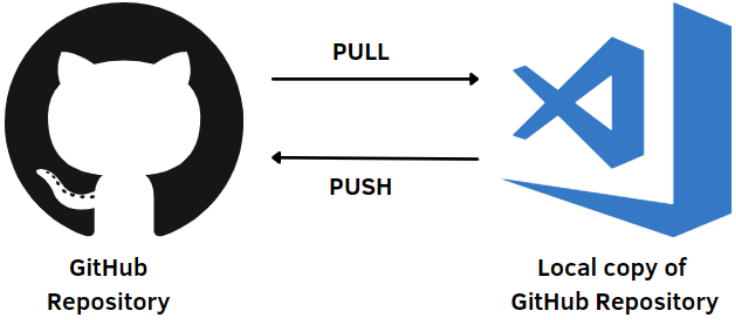
Creating GitHub Repository

1. Login
2. New Repository
3. Name it
4. Initialize with README



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Git Commands

```
git init # start tracking

git status # check changes

git add . # stage files

git commit -m "message" # save snapshot
```


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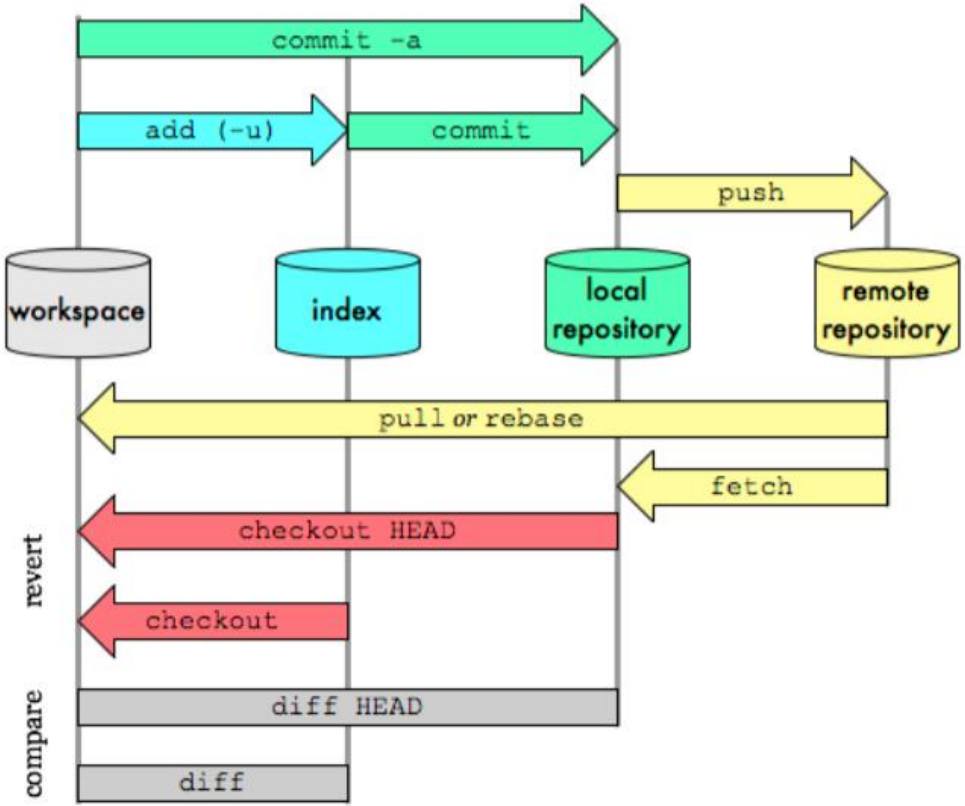
Git Workflow Diagram

- Working Directory
- Staging Area
- Repository



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First Python Program

```
print("Hello, Python")
```

Explanation:

- print() outputs text
- Quotes indicate string

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Running Python Program

- IDE Run
- Terminal execution

  Python file.py

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Variables

- Container for data
- Name → Value

Rules:

- No spaces
- Cannot start with number

Example:

name = "Anita"

age = 30

salary = 45000.75

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Python Data Types

Type	Description	Example
`int`	Whole numbers (no decimals)	`10`, `-5`
`float`	Decimal / floating-point numbers	`10.5`, `3.14`
`str`	Text (string values)	`"Hello"`
`bool`	Boolean (True or False)	`True`, `False`

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Type Checking & Casting

```
x = "100"
print(type(x))
y = int
```

Use Case: User input conversion

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Input & Output

```
name = input("Enter name: ")
age = int(input("Enter age: "))
print("Name:", name)
```


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Operators

- Arithmetic
- Comparison
- Logical
- Assignment

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Arithmetic Operators

```
a = 20

b = 6

print(a / b)

print(a // b)

print(a % b)
```


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Comparison Operators

```
print(a > b)

print(a == b)
```


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Logical Operators

```
age = 25

print(age > 18 and age < 60)
```


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Why Conditional Statements?

- Decision making
- Different outputs for different inputs

if Statement

```
balance = 5000

if balance > 1000:

print("Sufficient balance")
```


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Why Conditional Statements?

If–else

```
password = "admin123"
if password == "admin123":
print("Login success")
else:
print("Login failed")
```

If–elif–else

```
marks = 78
if marks >= 90:
grade = "A"
elif marks >= 75:
grade = "B"
elif marks >= 50:
grade = "C"
else:
grade = "Fail"
print(grade)
```


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Scenario Problem – Electricity Bill

Rules:

- First 100 units free
- Next 100 → ₹5/unit
- Above 200 → ₹10/unit

```
units = int(input("Units: "))
bill = 0
if units > 200:
bill = 500 + (units - 200) * 10
elif units > 100:
bill = (units - 100) * 5
print("Bill:", bill)
```


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Thank You!